Smart Medication Dosage Adjustment System

Closed-Loop Rate-Limited Infusion Control with Safety Failsafes

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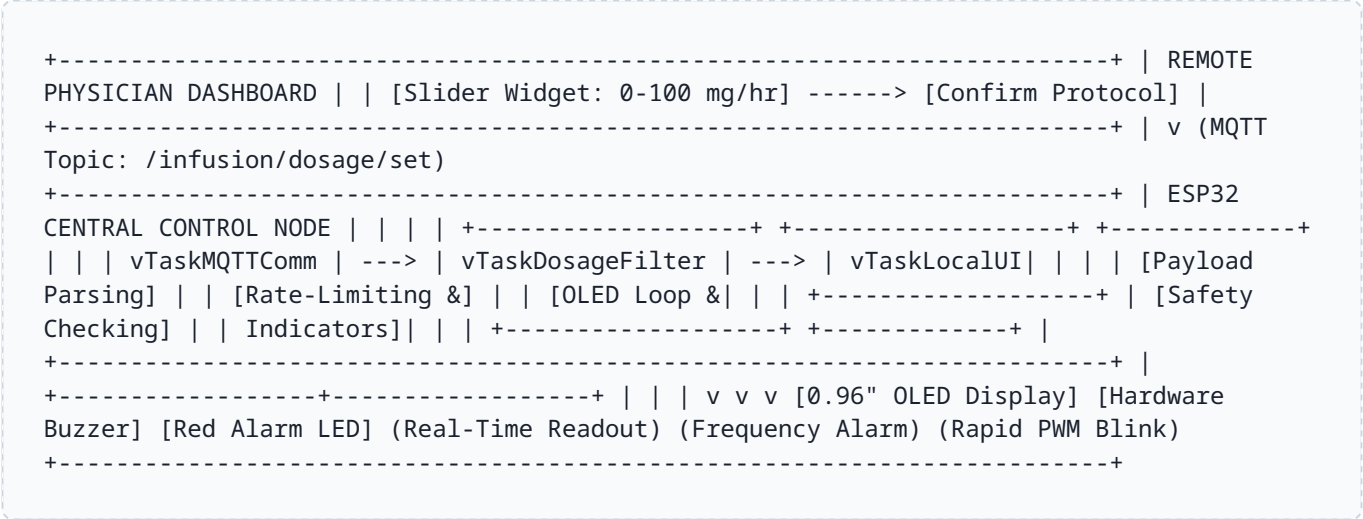
1. System Architecture Diagram

The system is designed around a secure, fault-tolerant network topology where remote physician console demands undergo client-side filtering, edge-level firmware rate-limiting, and physical output checking prior to actual infusion delivery. Local visual and audio indicators act as fail-safe alarms.



2. FreeRTOS Task Structure Explanation

To safely handle asynchronous user adjustments while continuously running hardware safety alarms, execution threads are isolated into deterministic FreeRTOS tasks governed by strict preemption policies.

Task Schedule Configurations

Task Handle Name	Priority	Execution Metric	Functional System Objective
vTaskMQTTComm	Medium (3)	200ms (Periodic)	Maintains cloud connection, intercepts remote control actions, and pushes performance telemetry back to logs.
vTaskDosageFilter	Highest (5)	50ms (Periodic)	Executes rate-limiting slopes, checks safety thresholds, and drives physical infusion metrics.
vTaskSafetyAlarms	High (4)	Event-Driven	Manages rapid LED blinking loops and sound alarms when threshold breaches occur.
vTaskLocalUI	Low (1)	200ms (Periodic)	Refreshes the local 0.96" OLED frame buffers with current rates and warning strings.
vTaskWatchdog	Highest (6)	100ms (Periodic)	Independent system supervisor that monitors for task execution locks or code loops.

Synchronization Primitives

- **Dosage State Mutex:** Encapsulates the current target dosage variable to prevent multi-task rewrite corruption between network adjustments and safety processing cycles.
- **Alarm Notification Flags:** Allows the filtration task to directly unblock the emergency alerting sequence without adding system scheduling overhead.

3. System Feed Mapping Table

Communication utilizes isolated topics to keep configuration instructions, hardware notifications, and telemetry updates cleanly separated.

MQTT Topic Namespace	Data Format	Operational Traffic Direction	Description / Structural Context
`/infusion/dosage/set`	Integer (0-100)	Subscribe (Inbound)	Target injection command requested via the remote dashboard slider component.
`/infusion/dosage/current`	Float (mg/hr)	Publish (Outbound)	Real-time, rate-limited dosage output level currently being delivered.
`/infusion/safety/level`	String Status	Publish (Outbound)	Current operational tier flag: `"NORMAL"`, `"WARNING"`, or `"CRITICAL"`.
`/infusion/safety/confirm`	Boolean (0/1)	Bi-Directional	Verification authorization token required to accept changes into critical dosage zones.

4. Remote UI Control Dashboard Design

The system control station provides real-time state feedback and confirmation locks to enforce clinical validation rules.

[Doctor Central Infusion Control Console Window]

• Dosage Command Slider: Main adjustment input widget bounded strictly to an operating range of 0 to 100 mg/hr.

• Safety State Visualizer: Tri-color status indicator (Green = Normal, Yellow = Warning, Red Flashing = Critical).

• Inline Overlay Dialog Box: Pops up automatically when values exceed 80 mg/hr, requiring an explicit confirmation click.

5. Multi-Tiered Safety Levels & Core Logic

The control firmware categorizes operating ranges into three explicit safety tiers based on delivery density:

Safety Tier State	Dosage Band Range	Local UI Indication Profile	System Operational Action
NORMAL	0 to 50 mg/hr	OLED shows standard rate text. Status indicator stays solid green.	Standard infusion operation. Smooth transitions allowed.
WARNING	51 to 80 mg/hr	OLED displays clear `[WARNING]` alert tag. Yellow status active.	System logs a warning flag. Rate-limiting strictly maintained.
CRITICAL	81 to 100 mg/hr	OLED flashes `!!CRITICAL!!`. Red LED blinks at 5Hz; buzzer sounds.	Restricts output until explicit user authorization code confirmation occurs.

6. Rate-Limiting Sloping Logic Explanation

To shield patients from dangerous sudden changes in medication delivery caused by accidental slider adjustments, the control task applies an internal mathematical slew-rate filter.

When a new target rate is requested, the system enforces a maximum change threshold ($\Delta R_{max} = 5 ext{ mg/hr/sec}$). This constraint smooths rapid step changes into a safe, linear adjustment curve:

$$R_{current}(t) = R_{current}(t-\Delta t) + ext{clamp}\left(R_{target} - R_{current}(t-\Delta t), \; -\Delta R_{max} \cdot \Delta t, \; \Delta R_{max} \cdot \Delta t ight)$$

This approach ensures that even if an input suddenly jumps from 20 mg/hr to 90 mg/hr, the actual delivery output escalates safely over several seconds, giving staff time to intervene if an error occurred.

```
// Slew-Rate Filtration Function Block
float apply_infusion_rate_limit(float target, float current, float delta_time) {
    float max_change = 5.0 * delta_time; // 5 mg/hr per second tracking rule
    float error = target - current;

    if (error > max_change)       current += max_change;
    else if (error < -max_change) current -= max_change;
    else                          current = target;

    return current;
}
```

7. Critical Zones Confirmation Sequence

When an incoming control command pushes past the 80 mg/hr threshold, a safety interlock sequence prevents accidental over-medication:

- **Command Interception:** The filtration logic catches the high adjustment command and holds the current delivery level at exactly 80 mg/hr.
- **Verification Request:** The device sends a confirmation prompt back to the control dashboard over the `/infusion/safety/confirm` channel, popping up an alert on the user interface.
- **Timeout and Rejection Rules:** If the physician confirms the change within 30 seconds, the system safely ramps up to the new target value. If the prompt times out or is rejected, the system clears the request and safely drops back to a baseline level of 50 mg/hr.

8. Local Hardware Fail-Safes & Diagnostics

In addition to network protections, local physical alarms ensure patient safety even during a network disconnect:

- **Independent Hardware Watchdog:** The dedicated `vTaskWatchdog` runs on an independent internal timer. If a firmware glitch or memory exception locks up the core filtering tasks, the watchdog timer will expire and automatically reset the system into a safe, zero-dosage state.
- **Direct Hardware Alerts:** The alert LED and buzzer lines bypass software queues and are driven directly by high-priority event routines, ensuring warning signals activate immediately even under heavy processor load.